

# Dataset Fairness Evaluation Report

## Stage 4: Base Fairness ML Model

### Detailed Group Metrics: Age

| Group       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged | 4174  | 0.7978   | 0.7746   | 0.3586    | 0.3208         | 0.3347 | 0.1281 |
| Young       | 5579  | 0.9215   | 0.7674   | 0.1129    | 0.0731         | 0.5238 | 0.0218 |
| Senior      | 2458  | 0.8072   | 0.7678   | 0.3234    | 0.2648         | 0.3887 | 0.0992 |

### Detailed Group Metrics: Marital-status

| Group                 | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Never-married         | 4065  | 0.9702   | 0.7551   | 0.0440    | 0.0187         | 0.6257 | 0.0023 |
| Married-civ-spouse    | 5567  | 0.7370   | 0.7307   | 0.4489    | 0.3975         | 0.3501 | 0.1920 |
| Divorced              | 1682  | 0.9310   | 0.7357   | 0.0957    | 0.0446         | 0.6273 | 0.0099 |
| Separated             | 373   | 0.9491   | 0.7656   | 0.0777    | 0.0375         | 0.5862 | 0.0058 |
| Married-spouse-absent | 152   | 0.9276   | 0.7764   | 0.1184    | 0.0592         | 0.5556 | 0.0075 |
| Widowed               | 359   | 0.9387   | 0.7088   | 0.0864    | 0.0251         | 0.7097 | 0.0000 |
| Married-AF-spouse     | 13    | 0.7692   | 0.7068   | 0.3846    | 0.1538         | 0.6000 | 0.0000 |

### Detailed Group Metrics: Occupation

| Group           | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------|-------|----------|----------|-----------|----------------|--------|--------|
| Other-service   | 1248  | 0.9583   | 0.6179   | 0.0441    | 0.0120         | 0.8364 | 0.0050 |
| Farming-fishing | 370   | 0.8865   | 0.6990   | 0.1243    | 0.0865         | 0.6087 | 0.0432 |

|                       |      |        |        |        |        |        |        |
|-----------------------|------|--------|--------|--------|--------|--------|--------|
| Sales                 | 1368 | 0.8363 | 0.7745 | 0.2617 | 0.2149 | 0.4022 | 0.0792 |
| Exec-manage<br>rial   | 1472 | 0.8003 | 0.8001 | 0.5061 | 0.4626 | 0.2403 | 0.1582 |
| Craft-repair          | 1564 | 0.8037 | 0.6870 | 0.2212 | 0.1682 | 0.5636 | 0.0920 |
| Prof-specialty        | 1568 | 0.8131 | 0.8072 | 0.4298 | 0.3948 | 0.2582 | 0.1331 |
| Machine-op-i<br>nspct | 727  | 0.8569 | 0.6223 | 0.1486 | 0.0633 | 0.7685 | 0.0339 |
| Tech-support          | 394  | 0.8223 | 0.7685 | 0.3020 | 0.2157 | 0.4370 | 0.0655 |
| Adm-clerical          | 1413 | 0.9002 | 0.7707 | 0.1331 | 0.1154 | 0.4415 | 0.0473 |
| Missing               | 715  | 0.9231 | 0.7269 | 0.0965 | 0.0559 | 0.6087 | 0.0201 |
| Handlers-clea<br>ners | 514  | 0.9455 | 0.7264 | 0.0700 | 0.0350 | 0.6389 | 0.0105 |
| Transport-mo<br>ving  | 576  | 0.8385 | 0.7032 | 0.1875 | 0.1372 | 0.5648 | 0.0684 |
| Protective-ser<br>v   | 229  | 0.7729 | 0.7181 | 0.2926 | 0.2664 | 0.4328 | 0.1420 |
| Priv-house-se<br>rv   | 52   | 0.9808 | 0.8949 | 0.0577 | 0.0385 | 0.3333 | 0.0000 |
| Armed-Force<br>s      | 1    | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |

### Detailed Group Metrics: Relationship

| Group          | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------|-------|----------|----------|-----------|----------------|--------|--------|
| Own-child      | 1904  | 0.9921   | 0.7707   | 0.0126    | 0.0047         | 0.6250 | 0.0000 |
| Husband        | 4916  | 0.7347   | 0.7289   | 0.4530    | 0.3997         | 0.3516 | 0.1938 |
| Not-in-family  | 3148  | 0.9327   | 0.7484   | 0.0985    | 0.0457         | 0.6097 | 0.0081 |
| Unmarried      | 1311  | 0.9519   | 0.7135   | 0.0648    | 0.0229         | 0.6941 | 0.0033 |
| Wife           | 580   | 0.7397   | 0.7357   | 0.4517    | 0.4259         | 0.3168 | 0.2138 |
| Other-relative | 352   | 0.9688   | 0.6685   | 0.0398    | 0.0085         | 0.7857 | 0.0000 |

### Detailed Group Metrics: Race

| Group | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------|-------|----------|----------|-----------|----------------|--------|--------|
| White | 10442 | 0.8476   | 0.7868   | 0.2547    | 0.2110         | 0.3850 | 0.0729 |
| Black | 1186  | 0.9250   | 0.7960   | 0.1214    | 0.0835         | 0.4653 | 0.0211 |

|                    |     |        |        |        |        |        |        |
|--------------------|-----|--------|--------|--------|--------|--------|--------|
| Amer-Indian-Eskimo | 126 | 0.9206 | 0.7697 | 0.0873 | 0.1032 | 0.3636 | 0.0522 |
| Asian-Pac-Islander | 369 | 0.8428 | 0.7802 | 0.2602 | 0.2060 | 0.4062 | 0.0696 |
| Other              | 88  | 0.9091 | 0.7525 | 0.1250 | 0.0795 | 0.5455 | 0.0260 |

### Detailed Group Metrics: Sex

| Group  | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------|-------|----------|----------|-----------|----------------|--------|--------|
| Female | 4057  | 0.9300   | 0.7967   | 0.1117    | 0.0786         | 0.4614 | 0.0208 |
| Male   | 8154  | 0.8195   | 0.7756   | 0.3028    | 0.2550         | 0.3771 | 0.0952 |

### Detailed Group Metrics: Native-country

| Group         | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------|-------|----------|----------|-----------|----------------|--------|--------|
| United-States | 10993 | 0.8543   | 0.7894   | 0.2430    | 0.2020         | 0.3841 | 0.0692 |
| Ireland       | 12    | 0.8333   | 0.8125   | 0.4167    | 0.2500         | 0.4000 | 0.0000 |
| Cuba          | 30    | 0.8667   | 0.8137   | 0.2667    | 0.2000         | 0.3750 | 0.0455 |
| Missing       | 209   | 0.8660   | 0.8208   | 0.2727    | 0.2249         | 0.3333 | 0.0592 |
| Puerto-Rico   | 49    | 0.9796   | 0.9231   | 0.0816    | 0.0612         | 0.2500 | 0.0000 |
| Philippines   | 73    | 0.8356   | 0.7349   | 0.2192    | 0.1644         | 0.5000 | 0.0702 |
| China         | 32    | 0.8750   | 0.8454   | 0.3125    | 0.2500         | 0.3000 | 0.0455 |
| Germany       | 54    | 0.8519   | 0.7974   | 0.2593    | 0.2222         | 0.3571 | 0.0750 |
| Mexico        | 211   | 0.9668   | 0.6026   | 0.0284    | 0.0142         | 0.8333 | 0.0098 |
| Peru          | 11    | 1.0000   | 1.0000   | 0.0909    | 0.0909         | 0.0000 | 0.0000 |
| India         | 36    | 0.7500   | 0.7402   | 0.4722    | 0.3333         | 0.4118 | 0.1053 |
| Guatemala     | 21    | 1.0000   | 1.0000   | 0.0000    | 0.0000         | 0      | 0.0000 |
| Taiwan        | 15    | 0.8667   | 0.8661   | 0.4667    | 0.4667         | 0.1429 | 0.1250 |
| Vietnam       | 23    | 0.9565   | 0.8217   | 0.0435    | 0.0870         | 0.0000 | 0.0455 |
| Jamaica       | 26    | 0.8077   | 0.4468   | 0.1923    | 0.0000         | 1.0000 | 0.0000 |
| Hungary       | 4     | 1.0000   | 1.0000   | 0.2500    | 0.2500         | 0.0000 | 0.0000 |
| Portugal      | 10    | 0.7000   | 0.4118   | 0.3000    | 0.0000         | 1.0000 | 0.0000 |
| Greece        | 14    | 0.5714   | 0.5625   | 0.5714    | 0.2857         | 0.6250 | 0.1667 |
| Haiti         | 20    | 0.9500   | 0.8857   | 0.1500    | 0.1000         | 0.3333 | 0.0000 |

|                                   |    |        |        |        |        |        |        |
|-----------------------------------|----|--------|--------|--------|--------|--------|--------|
| Nicaragua                         | 17 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| South                             | 26 | 0.8846 | 0.8462 | 0.2692 | 0.2308 | 0.2857 | 0.0526 |
| France                            | 7  | 0.4286 | 0.4167 | 0.4286 | 0.7143 | 0.3333 | 0.7500 |
| Canada                            | 48 | 0.7292 | 0.6485 | 0.3542 | 0.1667 | 0.6471 | 0.0645 |
| El-Salvador                       | 35 | 0.9429 | 0.8177 | 0.1143 | 0.0571 | 0.5000 | 0.0000 |
| England                           | 38 | 0.7632 | 0.7039 | 0.3421 | 0.2105 | 0.5385 | 0.0800 |
| Poland                            | 18 | 0.7222 | 0.5567 | 0.3333 | 0.0556 | 0.8333 | 0.0000 |
| Outlying-US(<br>Guam-USVI-<br>tc) | 9  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Honduras                          | 9  | 0.8889 | 0.4706 | 0.1111 | 0.0000 | 1.0000 | 0.0000 |
| Dominican-R<br>epublic            | 20 | 0.9500 | 0.8857 | 0.1000 | 0.1500 | 0.0000 | 0.0556 |
| Italy                             | 28 | 0.7857 | 0.7375 | 0.3571 | 0.2143 | 0.5000 | 0.0556 |
| Trinidad&To<br>bago               | 10 | 1.0000 | 1.0000 | 0.1000 | 0.1000 | 0.0000 | 0.0000 |
| Iran                              | 14 | 0.7143 | 0.5758 | 0.2143 | 0.2143 | 0.6667 | 0.1818 |
| Columbia                          | 19 | 0.9474 | 0.8190 | 0.0526 | 0.1053 | 0.0000 | 0.0556 |
| Hong                              | 7  | 0.7143 | 0.4167 | 0.1429 | 0.1429 | 1.0000 | 0.1667 |
| Yugoslavia                        | 7  | 0.7143 | 0.4167 | 0.2857 | 0.0000 | 1.0000 | 0.0000 |
| Ecuador                           | 7  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Cambodia                          | 7  | 0.5714 | 0.5333 | 0.5714 | 0.1429 | 0.7500 | 0.0000 |
| Japan                             | 23 | 0.8696 | 0.8392 | 0.3043 | 0.2609 | 0.2857 | 0.0625 |
| Scotland                          | 5  | 0.8000 | 0.4444 | 0.2000 | 0.0000 | 1.0000 | 0.0000 |
| Laos                              | 6  | 0.8333 | 0.7778 | 0.3333 | 0.1667 | 0.5000 | 0.0000 |
| Thailand                          | 8  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |

### Detailed Group Metrics: Education

| Group       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------------|-------|----------|----------|-----------|----------------|--------|--------|
| Bachelors   | 2071  | 0.8073   | 0.7999   | 0.4153    | 0.3916         | 0.2605 | 0.1445 |
| 9th         | 187   | 0.9679   | 0.4918   | 0.0321    | 0.0000         | 1.0000 | 0.0000 |
| HS-grad     | 4011  | 0.8679   | 0.6998   | 0.1576    | 0.0942         | 0.6203 | 0.0408 |
| Assoc-voc   | 497   | 0.8028   | 0.7245   | 0.2656    | 0.2012         | 0.4924 | 0.0904 |
| Prof-school | 224   | 0.8705   | 0.8304   | 0.7009    | 0.7857         | 0.0318 | 0.3582 |

|              |      |        |        |        |        |        |        |
|--------------|------|--------|--------|--------|--------|--------|--------|
| 10th         | 355  | 0.9380 | 0.6942 | 0.0789 | 0.0282 | 0.7143 | 0.0061 |
| Some-college | 2683 | 0.8580 | 0.7408 | 0.1826 | 0.1450 | 0.4918 | 0.0638 |
| 7th-8th      | 249  | 0.9197 | 0.5622 | 0.0884 | 0.0080 | 0.9091 | 0.0000 |
| Masters      | 637  | 0.7991 | 0.7977 | 0.5589 | 0.5243 | 0.2107 | 0.1886 |
| 11th         | 442  | 0.9615 | 0.6970 | 0.0475 | 0.0181 | 0.7143 | 0.0048 |
| 5th-6th      | 99   | 0.9091 | 0.4762 | 0.0909 | 0.0000 | 1.0000 | 0.0000 |
| Assoc-acdm   | 379  | 0.8311 | 0.7488 | 0.2427 | 0.1847 | 0.4674 | 0.0732 |
| Preschool    | 22   | 1.0000 | 1.0000 | 0.0455 | 0.0455 | 0.0000 | 0.0000 |
| 12th         | 154  | 0.9481 | 0.7640 | 0.0844 | 0.0325 | 0.6154 | 0.0000 |
| Doctorate    | 155  | 0.7161 | 0.6641 | 0.6581 | 0.7355 | 0.1569 | 0.5283 |
| 1st-4th      | 46   | 0.9783 | 0.4945 | 0.0217 | 0.0000 | 1.0000 | 0.0000 |

### Detailed Group Metrics: Race + Sex

| Group                       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White + Female              | 3261  | 0.9252   | 0.7997   | 0.1202    | 0.0883         | 0.4439 | 0.0244 |
| Black + Male                | 600   | 0.8900   | 0.7823   | 0.1783    | 0.1183         | 0.4766 | 0.0304 |
| White + Male                | 7181  | 0.8117   | 0.7732   | 0.3158    | 0.2721         | 0.3673 | 0.1056 |
| Black + Female              | 586   | 0.9573   | 0.7903   | 0.0631    | 0.0444         | 0.4865 | 0.0128 |
| Amer-Indian-Eskimo + Male   | 73    | 0.9178   | 0.7630   | 0.0685    | 0.1233         | 0.2000 | 0.0735 |
| Asian-Pac-Islander + Female | 131   | 0.9237   | 0.7707   | 0.1221    | 0.0611         | 0.5625 | 0.0087 |
| Amer-Indian-Eskimo + Female | 53    | 0.9057   | 0.6964   | 0.1132    | 0.0566         | 0.6667 | 0.0213 |
| Asian-Pac-Islander + Male   | 238   | 0.7815   | 0.7469   | 0.3361    | 0.2941         | 0.3875 | 0.1329 |
| Other + Female              | 26    | 0.9231   | 0.7292   | 0.0769    | 0.0769         | 0.5000 | 0.0417 |
| Other + Male                | 62    | 0.8871   | 0.6992   | 0.1452    | 0.0645         | 0.6667 | 0.0189 |

### Detailed Group Metrics: Age + Sex

| Group                | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged + Female | 1242  | 0.8961   | 0.8162   | 0.1892    | 0.1514         | 0.3745 | 0.0407 |
| Young + Male         | 3466  | 0.8987   | 0.7590   | 0.1437    | 0.0949         | 0.5221 | 0.0307 |
| Senior + Male        | 1756  | 0.7665   | 0.7519   | 0.4038    | 0.3536         | 0.3512 | 0.1538 |
| Young + Female       | 2113  | 0.9550   | 0.7672   | 0.0625    | 0.0393         | 0.5455 | 0.0116 |
| Middle-Aged + Male   | 2932  | 0.7538   | 0.7460   | 0.4304    | 0.3950         | 0.3273 | 0.1850 |
| Senior + Female      | 702   | 0.9117   | 0.7571   | 0.1225    | 0.0798         | 0.5349 | 0.0260 |

## Stage 4.5: Per-Attribute Fairness ML Model

### Detailed Group Metrics: Age

| Group       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged | 4174  | 0.7978   | 0.7746   | 0.3586    | 0.3208         | 0.3347 | 0.1281 |
| Young       | 5579  | 0.9215   | 0.7674   | 0.1129    | 0.0731         | 0.5238 | 0.0218 |
| Senior      | 2458  | 0.8072   | 0.7678   | 0.3234    | 0.2648         | 0.3887 | 0.0992 |

### Detailed Group Metrics: Marital-status

| Group                 | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Never-married         | 4065  | 0.9702   | 0.7551   | 0.0440    | 0.0187         | 0.6257 | 0.0023 |
| Married-civ-spouse    | 5567  | 0.7370   | 0.7307   | 0.4489    | 0.3975         | 0.3501 | 0.1920 |
| Divorced              | 1682  | 0.9310   | 0.7357   | 0.0957    | 0.0446         | 0.6273 | 0.0099 |
| Separated             | 373   | 0.9491   | 0.7656   | 0.0777    | 0.0375         | 0.5862 | 0.0058 |
| Married-spouse-absent | 152   | 0.9276   | 0.7764   | 0.1184    | 0.0592         | 0.5556 | 0.0075 |
| Widowed               | 359   | 0.9387   | 0.7088   | 0.0864    | 0.0251         | 0.7097 | 0.0000 |
| Married-AF-spouse     | 13    | 0.7692   | 0.7068   | 0.3846    | 0.1538         | 0.6000 | 0.0000 |

### Detailed Group Metrics: Occupation

| Group               | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Other-service       | 1248  | 0.9583   | 0.6179   | 0.0441    | 0.0120         | 0.8364 | 0.0050 |
| Farming-fishing     | 370   | 0.8865   | 0.6990   | 0.1243    | 0.0865         | 0.6087 | 0.0432 |
| Sales               | 1368  | 0.8363   | 0.7745   | 0.2617    | 0.2149         | 0.4022 | 0.0792 |
| Exec-managerial     | 1472  | 0.8003   | 0.8001   | 0.5061    | 0.4626         | 0.2403 | 0.1582 |
| Craft-repair        | 1564  | 0.8037   | 0.6870   | 0.2212    | 0.1682         | 0.5636 | 0.0920 |
| Prof-specialty      | 1568  | 0.8131   | 0.8072   | 0.4298    | 0.3948         | 0.2582 | 0.1331 |
| Machine-oper-inspct | 727   | 0.8569   | 0.6223   | 0.1486    | 0.0633         | 0.7685 | 0.0339 |
| Tech-support        | 394   | 0.8223   | 0.7685   | 0.3020    | 0.2157         | 0.4370 | 0.0655 |
| Adm-clerical        | 1413  | 0.9002   | 0.7707   | 0.1331    | 0.1154         | 0.4415 | 0.0473 |
| Missing             | 715   | 0.9231   | 0.7269   | 0.0965    | 0.0559         | 0.6087 | 0.0201 |
| Handlers-cleaners   | 514   | 0.9455   | 0.7264   | 0.0700    | 0.0350         | 0.6389 | 0.0105 |
| Transport-moving    | 576   | 0.8385   | 0.7032   | 0.1875    | 0.1372         | 0.5648 | 0.0684 |
| Protective-serv     | 229   | 0.7729   | 0.7181   | 0.2926    | 0.2664         | 0.4328 | 0.1420 |
| Priv-house-serv     | 52    | 0.9808   | 0.8949   | 0.0577    | 0.0385         | 0.3333 | 0.0000 |
| Armed-Forces        | 1     | 1.0000   | 1.0000   | 0.0000    | 0.0000         | 0      | 0.0000 |

### Detailed Group Metrics: Relationship

| Group          | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------|-------|----------|----------|-----------|----------------|--------|--------|
| Own-child      | 1904  | 0.9921   | 0.7707   | 0.0126    | 0.0047         | 0.6250 | 0.0000 |
| Husband        | 4916  | 0.7347   | 0.7289   | 0.4530    | 0.3997         | 0.3516 | 0.1938 |
| Not-in-family  | 3148  | 0.9327   | 0.7484   | 0.0985    | 0.0457         | 0.6097 | 0.0081 |
| Unmarried      | 1311  | 0.9519   | 0.7135   | 0.0648    | 0.0229         | 0.6941 | 0.0033 |
| Wife           | 580   | 0.7397   | 0.7357   | 0.4517    | 0.4259         | 0.3168 | 0.2138 |
| Other-relative | 352   | 0.9688   | 0.6685   | 0.0398    | 0.0085         | 0.7857 | 0.0000 |

### Detailed Group Metrics: Race

| Group              | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White              | 10442 | 0.8476   | 0.7868   | 0.2547    | 0.2110         | 0.3850 | 0.0729 |
| Black              | 1186  | 0.9250   | 0.7960   | 0.1214    | 0.0835         | 0.4653 | 0.0211 |
| Amer-Indian-Eskimo | 126   | 0.9206   | 0.7697   | 0.0873    | 0.1032         | 0.3636 | 0.0522 |
| Asian-Pac-Islander | 369   | 0.8428   | 0.7802   | 0.2602    | 0.2060         | 0.4062 | 0.0696 |
| Other              | 88    | 0.9091   | 0.7525   | 0.1250    | 0.0795         | 0.5455 | 0.0260 |

### Detailed Group Metrics: Sex

| Group  | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------|-------|----------|----------|-----------|----------------|--------|--------|
| Female | 4057  | 0.9300   | 0.7967   | 0.1117    | 0.0786         | 0.4614 | 0.0208 |
| Male   | 8154  | 0.8195   | 0.7756   | 0.3028    | 0.2550         | 0.3771 | 0.0952 |

### Detailed Group Metrics: Native-country

| Group         | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------|-------|----------|----------|-----------|----------------|--------|--------|
| United-States | 10993 | 0.8543   | 0.7894   | 0.2430    | 0.2020         | 0.3841 | 0.0692 |
| Ireland       | 12    | 0.8333   | 0.8125   | 0.4167    | 0.2500         | 0.4000 | 0.0000 |
| Cuba          | 30    | 0.8667   | 0.8137   | 0.2667    | 0.2000         | 0.3750 | 0.0455 |
| Missing       | 209   | 0.8660   | 0.8208   | 0.2727    | 0.2249         | 0.3333 | 0.0592 |
| Puerto-Rico   | 49    | 0.9796   | 0.9231   | 0.0816    | 0.0612         | 0.2500 | 0.0000 |
| Philippines   | 73    | 0.8356   | 0.7349   | 0.2192    | 0.1644         | 0.5000 | 0.0702 |
| China         | 32    | 0.8750   | 0.8454   | 0.3125    | 0.2500         | 0.3000 | 0.0455 |
| Germany       | 54    | 0.8519   | 0.7974   | 0.2593    | 0.2222         | 0.3571 | 0.0750 |
| Mexico        | 211   | 0.9668   | 0.6026   | 0.0284    | 0.0142         | 0.8333 | 0.0098 |
| Peru          | 11    | 1.0000   | 1.0000   | 0.0909    | 0.0909         | 0.0000 | 0.0000 |
| India         | 36    | 0.7500   | 0.7402   | 0.4722    | 0.3333         | 0.4118 | 0.1053 |
| Guatemala     | 21    | 1.0000   | 1.0000   | 0.0000    | 0.0000         | 0      | 0.0000 |
| Taiwan        | 15    | 0.8667   | 0.8661   | 0.4667    | 0.4667         | 0.1429 | 0.1250 |
| Vietnam       | 23    | 0.9565   | 0.8217   | 0.0435    | 0.0870         | 0.0000 | 0.0455 |
| Jamaica       | 26    | 0.8077   | 0.4468   | 0.1923    | 0.0000         | 1.0000 | 0.0000 |



|                                    |    |        |        |        |        |        |        |
|------------------------------------|----|--------|--------|--------|--------|--------|--------|
| Hungary                            | 4  | 1.0000 | 1.0000 | 0.2500 | 0.2500 | 0.0000 | 0.0000 |
| Portugal                           | 10 | 0.7000 | 0.4118 | 0.3000 | 0.0000 | 1.0000 | 0.0000 |
| Greece                             | 14 | 0.5714 | 0.5625 | 0.5714 | 0.2857 | 0.6250 | 0.1667 |
| Haiti                              | 20 | 0.9500 | 0.8857 | 0.1500 | 0.1000 | 0.3333 | 0.0000 |
| Nicaragua                          | 17 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| South                              | 26 | 0.8846 | 0.8462 | 0.2692 | 0.2308 | 0.2857 | 0.0526 |
| France                             | 7  | 0.4286 | 0.4167 | 0.4286 | 0.7143 | 0.3333 | 0.7500 |
| Canada                             | 48 | 0.7292 | 0.6485 | 0.3542 | 0.1667 | 0.6471 | 0.0645 |
| El-Salvador                        | 35 | 0.9429 | 0.8177 | 0.1143 | 0.0571 | 0.5000 | 0.0000 |
| England                            | 38 | 0.7632 | 0.7039 | 0.3421 | 0.2105 | 0.5385 | 0.0800 |
| Poland                             | 18 | 0.7222 | 0.5567 | 0.3333 | 0.0556 | 0.8333 | 0.0000 |
| Outlying-US(<br>Guam-USVI-e<br>tc) | 9  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Honduras                           | 9  | 0.8889 | 0.4706 | 0.1111 | 0.0000 | 1.0000 | 0.0000 |
| Dominican-R<br>epublic             | 20 | 0.9500 | 0.8857 | 0.1000 | 0.1500 | 0.0000 | 0.0556 |
| Italy                              | 28 | 0.7857 | 0.7375 | 0.3571 | 0.2143 | 0.5000 | 0.0556 |
| Trinidad&To<br>bago                | 10 | 1.0000 | 1.0000 | 0.1000 | 0.1000 | 0.0000 | 0.0000 |
| Iran                               | 14 | 0.7143 | 0.5758 | 0.2143 | 0.2143 | 0.6667 | 0.1818 |
| Columbia                           | 19 | 0.9474 | 0.8190 | 0.0526 | 0.1053 | 0.0000 | 0.0556 |
| Hong                               | 7  | 0.7143 | 0.4167 | 0.1429 | 0.1429 | 1.0000 | 0.1667 |
| Yugoslavia                         | 7  | 0.7143 | 0.4167 | 0.2857 | 0.0000 | 1.0000 | 0.0000 |
| Ecuador                            | 7  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Cambodia                           | 7  | 0.5714 | 0.5333 | 0.5714 | 0.1429 | 0.7500 | 0.0000 |
| Japan                              | 23 | 0.8696 | 0.8392 | 0.3043 | 0.2609 | 0.2857 | 0.0625 |
| Scotland                           | 5  | 0.8000 | 0.4444 | 0.2000 | 0.0000 | 1.0000 | 0.0000 |
| Laos                               | 6  | 0.8333 | 0.7778 | 0.3333 | 0.1667 | 0.5000 | 0.0000 |
| Thailand                           | 8  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |

### Detailed Group Metrics: Education

| Group     | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------|-------|----------|----------|-----------|----------------|--------|--------|
| Bachelors | 2071  | 0.8073   | 0.7999   | 0.4153    | 0.3916         | 0.2605 | 0.1445 |

|              |      |        |        |        |        |        |        |
|--------------|------|--------|--------|--------|--------|--------|--------|
| 9th          | 187  | 0.9679 | 0.4918 | 0.0321 | 0.0000 | 1.0000 | 0.0000 |
| HS-grad      | 4011 | 0.8679 | 0.6998 | 0.1576 | 0.0942 | 0.6203 | 0.0408 |
| Assoc-voc    | 497  | 0.8028 | 0.7245 | 0.2656 | 0.2012 | 0.4924 | 0.0904 |
| Prof-school  | 224  | 0.8705 | 0.8304 | 0.7009 | 0.7857 | 0.0318 | 0.3582 |
| 10th         | 355  | 0.9380 | 0.6942 | 0.0789 | 0.0282 | 0.7143 | 0.0061 |
| Some-college | 2683 | 0.8580 | 0.7408 | 0.1826 | 0.1450 | 0.4918 | 0.0638 |
| 7th-8th      | 249  | 0.9197 | 0.5622 | 0.0884 | 0.0080 | 0.9091 | 0.0000 |
| Masters      | 637  | 0.7991 | 0.7977 | 0.5589 | 0.5243 | 0.2107 | 0.1886 |
| 11th         | 442  | 0.9615 | 0.6970 | 0.0475 | 0.0181 | 0.7143 | 0.0048 |
| 5th-6th      | 99   | 0.9091 | 0.4762 | 0.0909 | 0.0000 | 1.0000 | 0.0000 |
| Assoc-acdm   | 379  | 0.8311 | 0.7488 | 0.2427 | 0.1847 | 0.4674 | 0.0732 |
| Preschool    | 22   | 1.0000 | 1.0000 | 0.0455 | 0.0455 | 0.0000 | 0.0000 |
| 12th         | 154  | 0.9481 | 0.7640 | 0.0844 | 0.0325 | 0.6154 | 0.0000 |
| Doctorate    | 155  | 0.7161 | 0.6641 | 0.6581 | 0.7355 | 0.1569 | 0.5283 |
| 1st-4th      | 46   | 0.9783 | 0.4945 | 0.0217 | 0.0000 | 1.0000 | 0.0000 |

## Stage 4.5: Intersectional Fairness ML Model

### Detailed Group Metrics: Race + Sex

| Group                       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White + Female              | 3261  | 0.9252   | 0.7997   | 0.1202    | 0.0883         | 0.4439 | 0.0244 |
| Black + Male                | 600   | 0.8900   | 0.7823   | 0.1783    | 0.1183         | 0.4766 | 0.0304 |
| White + Male                | 7181  | 0.8117   | 0.7732   | 0.3158    | 0.2721         | 0.3673 | 0.1056 |
| Black + Female              | 586   | 0.9573   | 0.7903   | 0.0631    | 0.0444         | 0.4865 | 0.0128 |
| Amer-Indian-Eskimo + Male   | 73    | 0.9178   | 0.7630   | 0.0685    | 0.1233         | 0.2000 | 0.0735 |
| Asian-Pac-Islander + Female | 131   | 0.9237   | 0.7707   | 0.1221    | 0.0611         | 0.5625 | 0.0087 |

|                             |     |        |        |        |        |        |        |
|-----------------------------|-----|--------|--------|--------|--------|--------|--------|
| Amer-Indian-Eskimo + Female | 53  | 0.9057 | 0.6964 | 0.1132 | 0.0566 | 0.6667 | 0.0213 |
| Asian-Pac-Islander + Male   | 238 | 0.7815 | 0.7469 | 0.3361 | 0.2941 | 0.3875 | 0.1329 |
| Other + Female              | 26  | 0.9231 | 0.7292 | 0.0769 | 0.0769 | 0.5000 | 0.0417 |
| Other + Male                | 62  | 0.8871 | 0.6992 | 0.1452 | 0.0645 | 0.6667 | 0.0189 |

Detailed Group Metrics: Age + Sex

| Group                | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged + Female | 1242  | 0.8961   | 0.8162   | 0.1892    | 0.1514         | 0.3745 | 0.0407 |
| Young + Male         | 3466  | 0.8987   | 0.7590   | 0.1437    | 0.0949         | 0.5221 | 0.0307 |
| Senior + Male        | 1756  | 0.7665   | 0.7519   | 0.4038    | 0.3536         | 0.3512 | 0.1538 |
| Young + Female       | 2113  | 0.9550   | 0.7672   | 0.0625    | 0.0393         | 0.5455 | 0.0116 |
| Middle-Aged + Male   | 2932  | 0.7538   | 0.7460   | 0.4304    | 0.3950         | 0.3273 | 0.1850 |
| Senior + Female      | 702   | 0.9117   | 0.7571   | 0.1225    | 0.0798         | 0.5349 | 0.0260 |

Stage 6: Post-Mitigation ML Model (Reweighting)

Detailed Group Metrics: Age

| Group       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged | 4174  | 0.7954   | 0.7722   | 0.3586    | 0.3222         | 0.3360 | 0.1311 |
| Young       | 5579  | 0.9206   | 0.7616   | 0.1129    | 0.0704         | 0.5397 | 0.0208 |
| Senior      | 2458  | 0.8137   | 0.7742   | 0.3234    | 0.2583         | 0.3887 | 0.0896 |

Detailed Group Metrics: Marital-status

| Group         | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------|-------|----------|----------|-----------|----------------|--------|--------|
| Never-married | 4065  | 0.9688   | 0.7369   | 0.0440    | 0.0172         | 0.6592 | 0.0023 |

|                           |      |        |        |        |        |        |        |
|---------------------------|------|--------|--------|--------|--------|--------|--------|
| Married-civ-s<br>pouse    | 5567 | 0.7397 | 0.7334 | 0.4489 | 0.3973 | 0.3473 | 0.1894 |
| Divorced                  | 1682 | 0.9287 | 0.7106 | 0.0957 | 0.0363 | 0.6832 | 0.0066 |
| Separated                 | 373  | 0.9464 | 0.7477 | 0.0777 | 0.0349 | 0.6207 | 0.0058 |
| Married-spou<br>se-absent | 152  | 0.9276 | 0.7603 | 0.1184 | 0.0461 | 0.6111 | 0.0000 |
| Widowed                   | 359  | 0.9331 | 0.6823 | 0.0864 | 0.0251 | 0.7419 | 0.0030 |
| Married-AF-s<br>pouse     | 13   | 0.6923 | 0.5667 | 0.3846 | 0.0769 | 0.8000 | 0.0000 |

### *Detailed Group Metrics: Occupation*

| Group                 | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Other-service         | 1248  | 0.9599   | 0.6518   | 0.0441    | 0.0152         | 0.7818 | 0.0059 |
| Farming-fishi<br>ng   | 370   | 0.8838   | 0.6953   | 0.1243    | 0.0892         | 0.6087 | 0.0463 |
| Sales                 | 1368  | 0.8377   | 0.7746   | 0.2617    | 0.2091         | 0.4106 | 0.0743 |
| Exec-manage<br>rial   | 1472  | 0.7969   | 0.7964   | 0.5061    | 0.4443         | 0.2617 | 0.1431 |
| Craft-repair          | 1564  | 0.7992   | 0.6842   | 0.2212    | 0.1752         | 0.5578 | 0.0993 |
| Prof-specialty        | 1568  | 0.8119   | 0.8054   | 0.4298    | 0.3884         | 0.2671 | 0.1286 |
| Machine-op-i<br>nspct | 727   | 0.8666   | 0.6649   | 0.1486    | 0.0757         | 0.6944 | 0.0355 |
| Tech-support          | 394   | 0.8299   | 0.7763   | 0.3020    | 0.2081         | 0.4370 | 0.0545 |
| Adm-clerical          | 1413  | 0.9038   | 0.7738   | 0.1331    | 0.1090         | 0.4521 | 0.0416 |
| Missing               | 715   | 0.9245   | 0.7384   | 0.0965    | 0.0601         | 0.5797 | 0.0217 |
| Handlers-clea<br>ners | 514   | 0.9455   | 0.7264   | 0.0700    | 0.0350         | 0.6389 | 0.0105 |
| Transport-mo<br>ving  | 576   | 0.8299   | 0.6885   | 0.1875    | 0.1389         | 0.5833 | 0.0748 |
| Protective-ser<br>v   | 229   | 0.7773   | 0.7274   | 0.2926    | 0.2795         | 0.4030 | 0.1481 |
| Priv-house-se<br>rv   | 52    | 0.9808   | 0.8949   | 0.0577    | 0.0385         | 0.3333 | 0.0000 |
| Armed-Force<br>s      | 1     | 1.0000   | 1.0000   | 0.0000    | 0.0000         | 0      | 0.0000 |

### *Detailed Group Metrics: Relationship*

| Group          | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------|-------|----------|----------|-----------|----------------|--------|--------|
| Own-child      | 1904  | 0.9921   | 0.7707   | 0.0126    | 0.0047         | 0.6250 | 0.0000 |
| Husband        | 4916  | 0.7362   | 0.7305   | 0.4530    | 0.4020         | 0.3476 | 0.1945 |
| Not-in-family  | 3148  | 0.9288   | 0.7228   | 0.0985    | 0.0394         | 0.6613 | 0.0067 |
| Unmarried      | 1311  | 0.9512   | 0.7015   | 0.0648    | 0.0206         | 0.7176 | 0.0024 |
| Wife           | 580   | 0.7517   | 0.7464   | 0.4517    | 0.4034         | 0.3282 | 0.1824 |
| Other-relative | 352   | 0.9688   | 0.6685   | 0.0398    | 0.0085         | 0.7857 | 0.0000 |

### Detailed Group Metrics: Race

| Group              | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White              | 10442 | 0.8479   | 0.7868   | 0.2547    | 0.2097         | 0.3868 | 0.0718 |
| Black              | 1186  | 0.9233   | 0.7867   | 0.1214    | 0.0784         | 0.4931 | 0.0192 |
| Amer-Indian-Eskimo | 126   | 0.9127   | 0.7143   | 0.0873    | 0.0794         | 0.5455 | 0.0435 |
| Asian-Pac-Islander | 369   | 0.8455   | 0.7831   | 0.2602    | 0.2033         | 0.4062 | 0.0659 |
| Other              | 88    | 0.9091   | 0.7250   | 0.1250    | 0.0568         | 0.6364 | 0.0130 |

### Detailed Group Metrics: Sex

| Group  | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------|-------|----------|----------|-----------|----------------|--------|--------|
| Female | 4057  | 0.9310   | 0.7953   | 0.1117    | 0.0742         | 0.4768 | 0.0178 |
| Male   | 8154  | 0.8191   | 0.7749   | 0.3028    | 0.2541         | 0.3791 | 0.0948 |

### Detailed Group Metrics: Native-country

| Group         | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------|-------|----------|----------|-----------|----------------|--------|--------|
| United-States | 10993 | 0.8543   | 0.7890   | 0.2430    | 0.2008         | 0.3867 | 0.0684 |
| Ireland       | 12    | 0.8333   | 0.8125   | 0.4167    | 0.2500         | 0.4000 | 0.0000 |
| Cuba          | 30    | 0.8667   | 0.8137   | 0.2667    | 0.2000         | 0.3750 | 0.0455 |
| Missing       | 209   | 0.8660   | 0.8184   | 0.2727    | 0.2153         | 0.3509 | 0.0526 |
| Puerto-Rico   | 49    | 0.9796   | 0.9231   | 0.0816    | 0.0612         | 0.2500 | 0.0000 |
| Philippines   | 73    | 0.8630   | 0.7902   | 0.2192    | 0.1918         | 0.3750 | 0.0702 |

|                                    |     |        |        |        |        |        |        |
|------------------------------------|-----|--------|--------|--------|--------|--------|--------|
| China                              | 32  | 0.7812 | 0.7196 | 0.3125 | 0.2188 | 0.5000 | 0.0909 |
| Germany                            | 54  | 0.8889 | 0.8480 | 0.2593 | 0.2222 | 0.2857 | 0.0500 |
| Mexico                             | 211 | 0.9621 | 0.5903 | 0.0284 | 0.0190 | 0.8333 | 0.0146 |
| Peru                               | 11  | 1.0000 | 1.0000 | 0.0909 | 0.0909 | 0.0000 | 0.0000 |
| India                              | 36  | 0.7500 | 0.7333 | 0.4722 | 0.2778 | 0.4706 | 0.0526 |
| Guatemala                          | 21  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Taiwan                             | 15  | 0.8000 | 0.7964 | 0.4667 | 0.4000 | 0.2857 | 0.1250 |
| Vietnam                            | 23  | 0.9565 | 0.8217 | 0.0435 | 0.0870 | 0.0000 | 0.0455 |
| Jamaica                            | 26  | 0.8462 | 0.6232 | 0.1923 | 0.0385 | 0.8000 | 0.0000 |
| Hungary                            | 4   | 0.7500 | 0.4286 | 0.2500 | 0.0000 | 1.0000 | 0.0000 |
| Portugal                           | 10  | 0.7000 | 0.4118 | 0.3000 | 0.0000 | 1.0000 | 0.0000 |
| Greece                             | 14  | 0.5714 | 0.5625 | 0.5714 | 0.2857 | 0.6250 | 0.1667 |
| Haiti                              | 20  | 0.9500 | 0.8857 | 0.1500 | 0.1000 | 0.3333 | 0.0000 |
| Nicaragua                          | 17  | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| South                              | 26  | 0.9231 | 0.8917 | 0.2692 | 0.1923 | 0.2857 | 0.0000 |
| France                             | 7   | 0.5714 | 0.5714 | 0.4286 | 0.5714 | 0.3333 | 0.5000 |
| Canada                             | 48  | 0.7083 | 0.5872 | 0.3542 | 0.1042 | 0.7647 | 0.0323 |
| El-Salvador                        | 35  | 0.9714 | 0.9206 | 0.1143 | 0.0857 | 0.2500 | 0.0000 |
| England                            | 38  | 0.7632 | 0.6842 | 0.3421 | 0.1579 | 0.6154 | 0.0400 |
| Poland                             | 18  | 0.7222 | 0.5567 | 0.3333 | 0.0556 | 0.8333 | 0.0000 |
| Outlying-US(<br>Guam-USVI-e<br>tc) | 9   | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |
| Honduras                           | 9   | 0.8889 | 0.4706 | 0.1111 | 0.0000 | 1.0000 | 0.0000 |
| Dominican-R<br>epublic             | 20  | 0.9500 | 0.8198 | 0.1000 | 0.0500 | 0.5000 | 0.0000 |
| Italy                              | 28  | 0.7500 | 0.6813 | 0.3571 | 0.1786 | 0.6000 | 0.0556 |
| Trinidad&To<br>bago                | 10  | 1.0000 | 1.0000 | 0.1000 | 0.1000 | 0.0000 | 0.0000 |
| Iran                               | 14  | 0.7143 | 0.5758 | 0.2143 | 0.2143 | 0.6667 | 0.1818 |
| Columbia                           | 19  | 0.9474 | 0.8190 | 0.0526 | 0.1053 | 0.0000 | 0.0556 |
| Hong                               | 7   | 0.7143 | 0.4167 | 0.1429 | 0.1429 | 1.0000 | 0.1667 |
| Yugoslavia                         | 7   | 0.7143 | 0.4167 | 0.2857 | 0.0000 | 1.0000 | 0.0000 |
| Ecuador                            | 7   | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0      | 0.0000 |

|          |    |        |        |        |        |        |        |
|----------|----|--------|--------|--------|--------|--------|--------|
| Cambodia | 7  | 0.7143 | 0.7083 | 0.5714 | 0.2857 | 0.5000 | 0.0000 |
| Japan    | 23 | 0.9130 | 0.8873 | 0.3043 | 0.2174 | 0.2857 | 0.0000 |
| Scotland | 5  | 0.8000 | 0.4444 | 0.2000 | 0.0000 | 1.0000 | 0.0000 |
| Laos     | 6  | 0.8333 | 0.7778 | 0.3333 | 0.1667 | 0.5000 | 0.0000 |
| Thailand | 8  | 0.8750 | 0.4667 | 0.0000 | 0.1250 | 0      | 0.1250 |

Detailed Group Metrics: Education

| Group        | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------|-------|----------|----------|-----------|----------------|--------|--------|
| Bachelors    | 2071  | 0.8112   | 0.8028   | 0.4153    | 0.3781         | 0.2721 | 0.1296 |
| 9th          | 187   | 0.9679   | 0.4918   | 0.0321    | 0.0000         | 1.0000 | 0.0000 |
| HS-grad      | 4011  | 0.8664   | 0.7044   | 0.1576    | 0.1022         | 0.5997 | 0.0465 |
| Assoc-voc    | 497   | 0.8028   | 0.7228   | 0.2656    | 0.1972         | 0.5000 | 0.0877 |
| Prof-school  | 224   | 0.8661   | 0.8374   | 0.7009    | 0.7188         | 0.0828 | 0.2537 |
| 10th         | 355   | 0.9352   | 0.6880   | 0.0789    | 0.0310         | 0.7143 | 0.0092 |
| Some-college | 2683  | 0.8595   | 0.7449   | 0.1826    | 0.1472         | 0.4816 | 0.0643 |
| 7th-8th      | 249   | 0.9197   | 0.5622   | 0.0884    | 0.0080         | 0.9091 | 0.0000 |
| Masters      | 637   | 0.8006   | 0.8001   | 0.5589    | 0.4914         | 0.2388 | 0.1495 |
| 11th         | 442   | 0.9593   | 0.7082   | 0.0475    | 0.0249         | 0.6667 | 0.0095 |
| 5th-6th      | 99    | 0.9091   | 0.4762   | 0.0909    | 0.0000         | 1.0000 | 0.0000 |
| Assoc-acdm   | 379   | 0.8285   | 0.7504   | 0.2427    | 0.1979         | 0.4457 | 0.0836 |
| Preschool    | 22    | 1.0000   | 1.0000   | 0.0455    | 0.0455         | 0.0000 | 0.0000 |
| 12th         | 154   | 0.9481   | 0.7640   | 0.0844    | 0.0325         | 0.6154 | 0.0000 |
| Doctorate    | 155   | 0.7032   | 0.6605   | 0.6581    | 0.6968         | 0.1961 | 0.4906 |
| 1st-4th      | 46    | 0.9783   | 0.4945   | 0.0217    | 0.0000         | 1.0000 | 0.0000 |

Detailed Group Metrics: Race + Sex

| Group          | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------|-------|----------|----------|-----------|----------------|--------|--------|
| White + Female | 3261  | 0.9264   | 0.7972   | 0.1202    | 0.0816         | 0.4668 | 0.0199 |
| Black + Male   | 600   | 0.8867   | 0.7736   | 0.1783    | 0.1150         | 0.4953 | 0.0304 |
| White + Male   | 7181  | 0.8123   | 0.7729   | 0.3158    | 0.2679         | 0.3730 | 0.1022 |

|                             |     |        |        |        |        |        |        |
|-----------------------------|-----|--------|--------|--------|--------|--------|--------|
| Black + Female              | 586 | 0.9608 | 0.8011 | 0.0631 | 0.0410 | 0.4865 | 0.0091 |
| Amer-Indian-Eskimo + Male   | 73  | 0.9041 | 0.7045 | 0.0685 | 0.1096 | 0.4000 | 0.0735 |
| Asian-Pac-Islander + Female | 131 | 0.9084 | 0.7248 | 0.1221 | 0.0611 | 0.6250 | 0.0174 |
| Amer-Indian-Eskimo + Female | 53  | 0.9245 | 0.7296 | 0.1132 | 0.0377 | 0.6667 | 0.0000 |
| Asian-Pac-Islander + Male   | 238 | 0.8109 | 0.7785 | 0.3361 | 0.2815 | 0.3625 | 0.1013 |
| Other + Female              | 26  | 0.9615 | 0.8231 | 0.0769 | 0.0385 | 0.5000 | 0.0000 |
| Other + Male                | 62  | 0.8871 | 0.6992 | 0.1452 | 0.0645 | 0.6667 | 0.0189 |

Detailed Group Metrics: Age + Sex

| Group                | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged + Female | 1242  | 0.8897   | 0.8011   | 0.1892    | 0.1433         | 0.4128 | 0.0397 |
| Young + Male         | 3466  | 0.8967   | 0.7508   | 0.1437    | 0.0912         | 0.5422 | 0.0296 |
| Senior + Male        | 1756  | 0.7722   | 0.7556   | 0.4038    | 0.3354         | 0.3667 | 0.1337 |
| Young + Female       | 2113  | 0.9598   | 0.7861   | 0.0625    | 0.0364         | 0.5303 | 0.0076 |
| Middle-Aged + Male   | 2932  | 0.7555   | 0.7480   | 0.4304    | 0.3980         | 0.3217 | 0.1862 |
| Senior + Female      | 702   | 0.9174   | 0.7575   | 0.1225    | 0.0655         | 0.5698 | 0.0146 |

Comparative Group Metrics: Before vs After (Reweighting)

Age

| Group       | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Middle-Aged | 0.7746      | 0.7722     | 0.3347       | 0.3360      | 0.1281       | 0.1311      | 0.3208             | 0.3222            |
| Young       | 0.7674      | 0.7616     | 0.5238       | 0.5397      | 0.0218       | 0.0208      | 0.0731             | 0.0704            |
| Senior      | 0.7678      | 0.7742     | 0.3887       | 0.3887      | 0.0992       | 0.0896      | 0.2648             | 0.2583            |

Marital-status



| Group                 | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-----------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Never-married         | 0.7551      | 0.7369     | 0.6257       | 0.6592      | 0.0023       | 0.0023      | 0.0187             | 0.0172            |
| Married-civ-spouse    | 0.7307      | 0.7334     | 0.3501       | 0.3473      | 0.1920       | 0.1894      | 0.3975             | 0.3973            |
| Divorced              | 0.7357      | 0.7106     | 0.6273       | 0.6832      | 0.0099       | 0.0066      | 0.0446             | 0.0363            |
| Separated             | 0.7656      | 0.7477     | 0.5862       | 0.6207      | 0.0058       | 0.0058      | 0.0375             | 0.0349            |
| Married-spouse-absent | 0.7764      | 0.7603     | 0.5556       | 0.6111      | 0.0075       | 0.0000      | 0.0592             | 0.0461            |
| Widowed               | 0.7088      | 0.6823     | 0.7097       | 0.7419      | 0.0000       | 0.0030      | 0.0251             | 0.0251            |
| Married-AF-spouse     | 0.7068      | 0.5667     | 0.6000       | 0.8000      | 0.0000       | 0.0000      | 0.1538             | 0.0769            |

## Occupation

| Group             | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Other-service     | 0.6179      | 0.6518     | 0.8364       | 0.7818      | 0.0050       | 0.0059      | 0.0120             | 0.0152            |
| Farming-fishing   | 0.6990      | 0.6953     | 0.6087       | 0.6087      | 0.0432       | 0.0463      | 0.0865             | 0.0892            |
| Sales             | 0.7745      | 0.7746     | 0.4022       | 0.4106      | 0.0792       | 0.0743      | 0.2149             | 0.2091            |
| Exec-managerial   | 0.8001      | 0.7964     | 0.2403       | 0.2617      | 0.1582       | 0.1431      | 0.4626             | 0.4443            |
| Craft-repair      | 0.6870      | 0.6842     | 0.5636       | 0.5578      | 0.0920       | 0.0993      | 0.1682             | 0.1752            |
| Prof-specialty    | 0.8072      | 0.8054     | 0.2582       | 0.2671      | 0.1331       | 0.1286      | 0.3948             | 0.3884            |
| Machine-op-inspct | 0.6223      | 0.6649     | 0.7685       | 0.6944      | 0.0339       | 0.0355      | 0.0633             | 0.0757            |
| Tech-support      | 0.7685      | 0.7763     | 0.4370       | 0.4370      | 0.0655       | 0.0545      | 0.2157             | 0.2081            |
| Adm-clerical      | 0.7707      | 0.7738     | 0.4415       | 0.4521      | 0.0473       | 0.0416      | 0.1154             | 0.1090            |
| Missing           | 0.7269      | 0.7384     | 0.6087       | 0.5797      | 0.0201       | 0.0217      | 0.0559             | 0.0601            |
| Handlers-cleaners | 0.7264      | 0.7264     | 0.6389       | 0.6389      | 0.0105       | 0.0105      | 0.0350             | 0.0350            |
| Transport-moving  | 0.7032      | 0.6885     | 0.5648       | 0.5833      | 0.0684       | 0.0748      | 0.1372             | 0.1389            |
| Protective-serv   | 0.7181      | 0.7274     | 0.4328       | 0.4030      | 0.1420       | 0.1481      | 0.2664             | 0.2795            |
| Priv-house-serv   | 0.8949      | 0.8949     | 0.3333       | 0.3333      | 0.0000       | 0.0000      | 0.0385             | 0.0385            |

|              |        |        |   |   |        |        |        |        |
|--------------|--------|--------|---|---|--------|--------|--------|--------|
| Armed-Forces | 1.0000 | 1.0000 | 0 | 0 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
|--------------|--------|--------|---|---|--------|--------|--------|--------|

### Relationship

| Group          | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|----------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Own-child      | 0.7707      | 0.7707     | 0.6250       | 0.6250      | 0.0000       | 0.0000      | 0.0047             | 0.0047            |
| Husband        | 0.7289      | 0.7305     | 0.3516       | 0.3476      | 0.1938       | 0.1945      | 0.3997             | 0.4020            |
| Not-in-family  | 0.7484      | 0.7228     | 0.6097       | 0.6613      | 0.0081       | 0.0067      | 0.0457             | 0.0394            |
| Unmarried      | 0.7135      | 0.7015     | 0.6941       | 0.7176      | 0.0033       | 0.0024      | 0.0229             | 0.0206            |
| Wife           | 0.7357      | 0.7464     | 0.3168       | 0.3282      | 0.2138       | 0.1824      | 0.4259             | 0.4034            |
| Other-relative | 0.6685      | 0.6685     | 0.7857       | 0.7857      | 0.0000       | 0.0000      | 0.0085             | 0.0085            |

### Race

| Group              | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| White              | 0.7868      | 0.7868     | 0.3850       | 0.3868      | 0.0729       | 0.0718      | 0.2110             | 0.2097            |
| Black              | 0.7960      | 0.7867     | 0.4653       | 0.4931      | 0.0211       | 0.0192      | 0.0835             | 0.0784            |
| Amer-Indian-Eskimo | 0.7697      | 0.7143     | 0.3636       | 0.5455      | 0.0522       | 0.0435      | 0.1032             | 0.0794            |
| Asian-Pac-Islander | 0.7802      | 0.7831     | 0.4062       | 0.4062      | 0.0696       | 0.0659      | 0.2060             | 0.2033            |
| Other              | 0.7525      | 0.7250     | 0.5455       | 0.6364      | 0.0260       | 0.0130      | 0.0795             | 0.0568            |

### Sex

| Group  | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Female | 0.7967      | 0.7953     | 0.4614       | 0.4768      | 0.0208       | 0.0178      | 0.0786             | 0.0742            |
| Male   | 0.7756      | 0.7749     | 0.3771       | 0.3791      | 0.0952       | 0.0948      | 0.2550             | 0.2541            |

### Native-country

| Group         | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|---------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| United-States | 0.7894      | 0.7890     | 0.3841       | 0.3867      | 0.0692       | 0.0684      | 0.2020             | 0.2008            |
| Ireland       | 0.8125      | 0.8125     | 0.4000       | 0.4000      | 0.0000       | 0.0000      | 0.2500             | 0.2500            |

|                                    |        |        |        |        |        |        |        |        |
|------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Cuba                               | 0.8137 | 0.8137 | 0.3750 | 0.3750 | 0.0455 | 0.0455 | 0.2000 | 0.2000 |
| Missing                            | 0.8208 | 0.8184 | 0.3333 | 0.3509 | 0.0592 | 0.0526 | 0.2249 | 0.2153 |
| Puerto-Rico                        | 0.9231 | 0.9231 | 0.2500 | 0.2500 | 0.0000 | 0.0000 | 0.0612 | 0.0612 |
| Philippines                        | 0.7349 | 0.7902 | 0.5000 | 0.3750 | 0.0702 | 0.0702 | 0.1644 | 0.1918 |
| China                              | 0.8454 | 0.7196 | 0.3000 | 0.5000 | 0.0455 | 0.0909 | 0.2500 | 0.2188 |
| Germany                            | 0.7974 | 0.8480 | 0.3571 | 0.2857 | 0.0750 | 0.0500 | 0.2222 | 0.2222 |
| Mexico                             | 0.6026 | 0.5903 | 0.8333 | 0.8333 | 0.0098 | 0.0146 | 0.0142 | 0.0190 |
| Peru                               | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0909 | 0.0909 |
| India                              | 0.7402 | 0.7333 | 0.4118 | 0.4706 | 0.1053 | 0.0526 | 0.3333 | 0.2778 |
| Guatemala                          | 1.0000 | 1.0000 | 0      | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Taiwan                             | 0.8661 | 0.7964 | 0.1429 | 0.2857 | 0.1250 | 0.1250 | 0.4667 | 0.4000 |
| Vietnam                            | 0.8217 | 0.8217 | 0.0000 | 0.0000 | 0.0455 | 0.0455 | 0.0870 | 0.0870 |
| Jamaica                            | 0.4468 | 0.6232 | 1.0000 | 0.8000 | 0.0000 | 0.0000 | 0.0000 | 0.0385 |
| Hungary                            | 1.0000 | 0.4286 | 0.0000 | 1.0000 | 0.0000 | 0.0000 | 0.2500 | 0.0000 |
| Portugal                           | 0.4118 | 0.4118 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Greece                             | 0.5625 | 0.5625 | 0.6250 | 0.6250 | 0.1667 | 0.1667 | 0.2857 | 0.2857 |
| Haiti                              | 0.8857 | 0.8857 | 0.3333 | 0.3333 | 0.0000 | 0.0000 | 0.1000 | 0.1000 |
| Nicaragua                          | 1.0000 | 1.0000 | 0      | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| South                              | 0.8462 | 0.8917 | 0.2857 | 0.2857 | 0.0526 | 0.0000 | 0.2308 | 0.1923 |
| France                             | 0.4167 | 0.5714 | 0.3333 | 0.3333 | 0.7500 | 0.5000 | 0.7143 | 0.5714 |
| Canada                             | 0.6485 | 0.5872 | 0.6471 | 0.7647 | 0.0645 | 0.0323 | 0.1667 | 0.1042 |
| El-Salvador                        | 0.8177 | 0.9206 | 0.5000 | 0.2500 | 0.0000 | 0.0000 | 0.0571 | 0.0857 |
| England                            | 0.7039 | 0.6842 | 0.5385 | 0.6154 | 0.0800 | 0.0400 | 0.2105 | 0.1579 |
| Poland                             | 0.5567 | 0.5567 | 0.8333 | 0.8333 | 0.0000 | 0.0000 | 0.0556 | 0.0556 |
| Outlying-US<br>(Guam-USV<br>I-etc) | 1.0000 | 1.0000 | 0      | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Honduras                           | 0.4706 | 0.4706 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Dominican-<br>Republic             | 0.8857 | 0.8198 | 0.0000 | 0.5000 | 0.0556 | 0.0000 | 0.1500 | 0.0500 |
| Italy                              | 0.7375 | 0.6813 | 0.5000 | 0.6000 | 0.0556 | 0.0556 | 0.2143 | 0.1786 |
| Trinidad&T<br>obago                | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.1000 | 0.1000 |
| Iran                               | 0.5758 | 0.5758 | 0.6667 | 0.6667 | 0.1818 | 0.1818 | 0.2143 | 0.2143 |

|            |        |        |        |        |        |        |        |        |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Columbia   | 0.8190 | 0.8190 | 0.0000 | 0.0000 | 0.0556 | 0.0556 | 0.1053 | 0.1053 |
| Hong       | 0.4167 | 0.4167 | 1.0000 | 1.0000 | 0.1667 | 0.1667 | 0.1429 | 0.1429 |
| Yugoslavia | 0.4167 | 0.4167 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Ecuador    | 1.0000 | 1.0000 | 0      | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Cambodia   | 0.5333 | 0.7083 | 0.7500 | 0.5000 | 0.0000 | 0.0000 | 0.1429 | 0.2857 |
| Japan      | 0.8392 | 0.8873 | 0.2857 | 0.2857 | 0.0625 | 0.0000 | 0.2609 | 0.2174 |
| Scotland   | 0.4444 | 0.4444 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Laos       | 0.7778 | 0.7778 | 0.5000 | 0.5000 | 0.0000 | 0.0000 | 0.1667 | 0.1667 |
| Thailand   | 1.0000 | 0.4667 | 0      | 0      | 0.0000 | 0.1250 | 0.0000 | 0.1250 |

### Education

| Group        | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Bachelors    | 0.7999      | 0.8028     | 0.2605       | 0.2721      | 0.1445       | 0.1296      | 0.3916             | 0.3781            |
| 9th          | 0.4918      | 0.4918     | 1.0000       | 1.0000      | 0.0000       | 0.0000      | 0.0000             | 0.0000            |
| HS-grad      | 0.6998      | 0.7044     | 0.6203       | 0.5997      | 0.0408       | 0.0465      | 0.0942             | 0.1022            |
| Assoc-voc    | 0.7245      | 0.7228     | 0.4924       | 0.5000      | 0.0904       | 0.0877      | 0.2012             | 0.1972            |
| Prof-school  | 0.8304      | 0.8374     | 0.0318       | 0.0828      | 0.3582       | 0.2537      | 0.7857             | 0.7188            |
| 10th         | 0.6942      | 0.6880     | 0.7143       | 0.7143      | 0.0061       | 0.0092      | 0.0282             | 0.0310            |
| Some-college | 0.7408      | 0.7449     | 0.4918       | 0.4816      | 0.0638       | 0.0643      | 0.1450             | 0.1472            |
| 7th-8th      | 0.5622      | 0.5622     | 0.9091       | 0.9091      | 0.0000       | 0.0000      | 0.0080             | 0.0080            |
| Masters      | 0.7977      | 0.8001     | 0.2107       | 0.2388      | 0.1886       | 0.1495      | 0.5243             | 0.4914            |
| 11th         | 0.6970      | 0.7082     | 0.7143       | 0.6667      | 0.0048       | 0.0095      | 0.0181             | 0.0249            |
| 5th-6th      | 0.4762      | 0.4762     | 1.0000       | 1.0000      | 0.0000       | 0.0000      | 0.0000             | 0.0000            |
| Assoc-acdm   | 0.7488      | 0.7504     | 0.4674       | 0.4457      | 0.0732       | 0.0836      | 0.1847             | 0.1979            |
| Preschool    | 1.0000      | 1.0000     | 0.0000       | 0.0000      | 0.0000       | 0.0000      | 0.0455             | 0.0455            |
| 12th         | 0.7640      | 0.7640     | 0.6154       | 0.6154      | 0.0000       | 0.0000      | 0.0325             | 0.0325            |
| Doctorate    | 0.6641      | 0.6605     | 0.1569       | 0.1961      | 0.5283       | 0.4906      | 0.7355             | 0.6968            |
| 1st-4th      | 0.4945      | 0.4945     | 1.0000       | 1.0000      | 0.0000       | 0.0000      | 0.0000             | 0.0000            |

### Race + Sex

| Group | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
|-------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|

|                             |        |        |        |        |        |        |        |        |
|-----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| White + Female              | 0.7997 | 0.7972 | 0.4439 | 0.4668 | 0.0244 | 0.0199 | 0.0883 | 0.0816 |
| Black + Male                | 0.7823 | 0.7736 | 0.4766 | 0.4953 | 0.0304 | 0.0304 | 0.1183 | 0.1150 |
| White + Male                | 0.7732 | 0.7729 | 0.3673 | 0.3730 | 0.1056 | 0.1022 | 0.2721 | 0.2679 |
| Black + Female              | 0.7903 | 0.8011 | 0.4865 | 0.4865 | 0.0128 | 0.0091 | 0.0444 | 0.0410 |
| Amer-Indian-Eskimo + Male   | 0.7630 | 0.7045 | 0.2000 | 0.4000 | 0.0735 | 0.0735 | 0.1233 | 0.1096 |
| Asian-Pac-Islander + Female | 0.7707 | 0.7248 | 0.5625 | 0.6250 | 0.0087 | 0.0174 | 0.0611 | 0.0611 |
| Amer-Indian-Eskimo + Female | 0.6964 | 0.7296 | 0.6667 | 0.6667 | 0.0213 | 0.0000 | 0.0566 | 0.0377 |
| Asian-Pac-Islander + Male   | 0.7469 | 0.7785 | 0.3875 | 0.3625 | 0.1329 | 0.1013 | 0.2941 | 0.2815 |
| Other + Female              | 0.7292 | 0.8231 | 0.5000 | 0.5000 | 0.0417 | 0.0000 | 0.0769 | 0.0385 |
| Other + Male                | 0.6992 | 0.6992 | 0.6667 | 0.6667 | 0.0189 | 0.0189 | 0.0645 | 0.0645 |

### Age + Sex

| Group                | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|----------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Middle-Aged + Female | 0.8162      | 0.8011     | 0.3745       | 0.4128      | 0.0407       | 0.0397      | 0.1514             | 0.1433            |
| Young + Male         | 0.7590      | 0.7508     | 0.5221       | 0.5422      | 0.0307       | 0.0296      | 0.0949             | 0.0912            |
| Senior + Male        | 0.7519      | 0.7556     | 0.3512       | 0.3667      | 0.1538       | 0.1337      | 0.3536             | 0.3354            |
| Young + Female       | 0.7672      | 0.7861     | 0.5455       | 0.5303      | 0.0116       | 0.0076      | 0.0393             | 0.0364            |
| Middle-Aged + Male   | 0.7460      | 0.7480     | 0.3273       | 0.3217      | 0.1850       | 0.1862      | 0.3950             | 0.3980            |
| Senior + Female      | 0.7571      | 0.7575     | 0.5349       | 0.5698      | 0.0260       | 0.0146      | 0.0798             | 0.0655            |

## Stage 6: Post-Mitigation ML Model (SMOTE)

### Detailed Group Metrics: Age

| Group       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged | 8245  | 0.8861   | 0.8686   | 0.6753    | 0.6900         | 0.0735 | 0.1980 |
| Senior      | 4321  | 0.8766   | 0.8702   | 0.6151    | 0.6075         | 0.1065 | 0.1503 |
| Young       | 6012  | 0.9173   | 0.8477   | 0.1768    | 0.1470         | 0.3180 | 0.0321 |

### Detailed Group Metrics: Marital-status

| Group                     | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|---------------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Married-civ-s<br>pouse    | 10737 | 0.8514   | 0.8154   | 0.7143    | 0.7271         | 0.0951 | 0.2826 |
| Divorced                  | 2043  | 0.9290   | 0.9015   | 0.2555    | 0.2159         | 0.2165 | 0.0210 |
| Never-marrie<br>d         | 4176  | 0.9648   | 0.8319   | 0.0694    | 0.0414         | 0.4552 | 0.0039 |
| Married-spou<br>se-absent | 519   | 0.9441   | 0.9258   | 0.7418    | 0.7553         | 0.0286 | 0.1343 |
| Widowed                   | 357   | 0.9356   | 0.7024   | 0.0812    | 0.0336         | 0.6897 | 0.0091 |
| Separated                 | 408   | 0.9387   | 0.8697   | 0.1569    | 0.1152         | 0.3281 | 0.0116 |
| Married-AF-s<br>pouse     | 338   | 0.9882   | 0.8970   | 0.9763    | 0.9645         | 0.0121 | 0.0000 |

### Detailed Group Metrics: Occupation

| Group                 | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Prof-specialty        | 2508  | 0.8780   | 0.8666   | 0.6435    | 0.6491         | 0.0905 | 0.1790 |
| Machine-op-i<br>nspct | 1189  | 0.9167   | 0.9166   | 0.4794    | 0.4836         | 0.0825 | 0.0840 |
| Handlers-clea<br>ners | 978   | 0.9550   | 0.9550   | 0.5112    | 0.5051         | 0.0500 | 0.0397 |
| Missing               | 788   | 0.9213   | 0.8697   | 0.1802    | 0.1904         | 0.1901 | 0.0542 |
| Protective-ser<br>v   | 714   | 0.8936   | 0.8515   | 0.7731    | 0.7591         | 0.0779 | 0.2037 |
| Priv-house-se<br>rv   | 509   | 0.9921   | 0.9778   | 0.9037    | 0.8998         | 0.0065 | 0.0204 |
| Exec-manage<br>rial   | 2323  | 0.8674   | 0.8458   | 0.6870    | 0.6870         | 0.0965 | 0.2118 |
| Other-service         | 1694  | 0.9575   | 0.9491   | 0.2957    | 0.2981         | 0.0679 | 0.0319 |

|                  |      |        |        |        |        |        |        |
|------------------|------|--------|--------|--------|--------|--------|--------|
| Farming-fishing  | 835  | 0.9329 | 0.9292 | 0.6120 | 0.6168 | 0.0509 | 0.0926 |
| Adm-clerical     | 1678 | 0.9088 | 0.8843 | 0.2700 | 0.2694 | 0.1700 | 0.0620 |
| Armed-Forces     | 195  | 0.9590 | 0.5895 | 0.9949 | 0.9538 | 0.0412 | 0.0000 |
| Craft-repair     | 2338 | 0.8426 | 0.8421 | 0.4790 | 0.4636 | 0.1804 | 0.1363 |
| Sales            | 1694 | 0.8636 | 0.8579 | 0.4038 | 0.3961 | 0.1784 | 0.1079 |
| Tech-support     | 521  | 0.8810 | 0.8800 | 0.4722 | 0.4376 | 0.1626 | 0.0800 |
| Transport-moving | 614  | 0.8046 | 0.7072 | 0.2378 | 0.1857 | 0.5205 | 0.0940 |

### Detailed Group Metrics: Relationship

| Group          | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|----------------|-------|----------|----------|-----------|----------------|--------|--------|
| Husband        | 9926  | 0.8563   | 0.8154   | 0.7291    | 0.7420         | 0.0897 | 0.2890 |
| Wife           | 668   | 0.7710   | 0.7702   | 0.5240    | 0.5344         | 0.2086 | 0.2516 |
| Other-relative | 654   | 0.9572   | 0.9571   | 0.4832    | 0.4618         | 0.0665 | 0.0207 |
| Unmarried      | 1521  | 0.9606   | 0.9340   | 0.1940    | 0.1716         | 0.1593 | 0.0106 |
| Not-in-family  | 3615  | 0.9239   | 0.8775   | 0.2149    | 0.1693         | 0.2831 | 0.0194 |
| Own-child      | 2194  | 0.9877   | 0.9745   | 0.1431    | 0.1372         | 0.0637 | 0.0037 |

### Detailed Group Metrics: Race

| Group              | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White              | 15672 | 0.8885   | 0.8885   | 0.5034    | 0.4991         | 0.1151 | 0.1079 |
| Black              | 1633  | 0.9357   | 0.9300   | 0.3619    | 0.3527         | 0.1015 | 0.0432 |
| Other              | 552   | 0.9656   | 0.9323   | 0.8605    | 0.8406         | 0.0316 | 0.0519 |
| Asian-Pac-Islander | 563   | 0.8579   | 0.8577   | 0.5151    | 0.5187         | 0.1345 | 0.1502 |
| Amer-Indian-Eskimo | 158   | 0.8924   | 0.8652   | 0.2722    | 0.2785         | 0.1860 | 0.0783 |

### Detailed Group Metrics: Sex

| Group | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-------|-------|----------|----------|-----------|----------------|--------|--------|
| Male  | 12866 | 0.8724   | 0.8707   | 0.5581    | 0.5574         | 0.1150 | 0.1435 |

|        |      |        |        |        |        |        |        |
|--------|------|--------|--------|--------|--------|--------|--------|
| Female | 5712 | 0.9428 | 0.9380 | 0.3690 | 0.3549 | 0.0968 | 0.0341 |
|--------|------|--------|--------|--------|--------|--------|--------|

### Detailed Group Metrics: Native-country

| Group              | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------------|-------|----------|----------|-----------|----------------|--------|--------|
| United-States      | 16322 | 0.8905   | 0.8905   | 0.4901    | 0.4859         | 0.1160 | 0.1032 |
| Puerto-Rico        | 100   | 0.9600   | 0.9597   | 0.5500    | 0.5300         | 0.0545 | 0.0222 |
| Yugoslavia         | 6     | 0.3333   | 0.2500   | 0.1667    | 0.5000         | 1.0000 | 0.6000 |
| Iran               | 45    | 0.8222   | 0.7727   | 0.7556    | 0.7111         | 0.1471 | 0.2727 |
| Germany            | 80    | 0.8875   | 0.8875   | 0.5000    | 0.4875         | 0.1250 | 0.1000 |
| Cuba               | 37    | 0.8919   | 0.8879   | 0.4054    | 0.4054         | 0.1333 | 0.0909 |
| Trinidad&Tobago    | 87    | 0.9425   | 0.8523   | 0.8966    | 0.8851         | 0.0385 | 0.2222 |
| Missing            | 219   | 0.8447   | 0.8157   | 0.3059    | 0.2968         | 0.2687 | 0.1053 |
| Greece             | 28    | 0.8929   | 0.8303   | 0.7857    | 0.8214         | 0.0455 | 0.3333 |
| Poland             | 63    | 0.9841   | 0.9734   | 0.8095    | 0.8254         | 0.0000 | 0.0833 |
| Laos               | 39    | 0.9744   | 0.9372   | 0.8974    | 0.8718         | 0.0286 | 0.0000 |
| South              | 69    | 0.9710   | 0.9637   | 0.7246    | 0.7246         | 0.0200 | 0.0526 |
| Dominican-Republic | 32    | 0.8750   | 0.8730   | 0.4375    | 0.4375         | 0.1429 | 0.1111 |
| Haiti              | 49    | 0.9184   | 0.9099   | 0.6531    | 0.6531         | 0.0625 | 0.1176 |
| Honduras           | 27    | 1.0000   | 1.0000   | 0.7037    | 0.7037         | 0.0000 | 0.0000 |
| Mexico             | 252   | 0.9563   | 0.9262   | 0.1865    | 0.1746         | 0.1489 | 0.0195 |
| Jamaica            | 56    | 0.9464   | 0.9443   | 0.6250    | 0.5714         | 0.0857 | 0.0000 |
| Taiwan             | 74    | 0.9595   | 0.9003   | 0.8919    | 0.8784         | 0.0303 | 0.1250 |
| Canada             | 60    | 0.9167   | 0.9165   | 0.4833    | 0.4667         | 0.1034 | 0.0645 |
| Columbia           | 32    | 1.0000   | 1.0000   | 0.4375    | 0.4375         | 0.0000 | 0.0000 |
| China              | 37    | 0.7838   | 0.7798   | 0.4054    | 0.4595         | 0.2000 | 0.2273 |
| Peru               | 47    | 1.0000   | 1.0000   | 0.7872    | 0.7872         | 0.0000 | 0.0000 |
| England            | 50    | 0.8600   | 0.8586   | 0.5000    | 0.4000         | 0.2400 | 0.0400 |
| Portugal           | 55    | 0.9455   | 0.8956   | 0.8727    | 0.8182         | 0.0625 | 0.0000 |
| Philippines        | 120   | 0.8667   | 0.8661   | 0.5250    | 0.5417         | 0.1111 | 0.1579 |
| Vietnam            | 29    | 0.8621   | 0.8389   | 0.2414    | 0.3793         | 0.0000 | 0.1818 |



|                             |    |        |        |        |        |        |        |
|-----------------------------|----|--------|--------|--------|--------|--------|--------|
| Outlying-US( Guam-USVI-etc) | 34 | 1.0000 | 1.0000 | 0.7353 | 0.7353 | 0.0000 | 0.0000 |
| Scotland                    | 56 | 0.9821 | 0.9396 | 0.9286 | 0.9107 | 0.0192 | 0.0000 |
| Nicaragua                   | 46 | 0.9565 | 0.9544 | 0.6304 | 0.5870 | 0.0690 | 0.0000 |
| Ireland                     | 34 | 0.9118 | 0.8717 | 0.7941 | 0.7647 | 0.0741 | 0.1429 |
| Hungary                     | 23 | 0.9130 | 0.8083 | 0.8696 | 0.8696 | 0.0500 | 0.3333 |
| Italy                       | 45 | 0.9111 | 0.9089 | 0.6000 | 0.5556 | 0.1111 | 0.0556 |
| India                       | 55 | 0.8000 | 0.7815 | 0.6545 | 0.6364 | 0.1667 | 0.2632 |
| Thailand                    | 68 | 0.9706 | 0.9292 | 0.8824 | 0.8824 | 0.0167 | 0.1250 |
| El-Salvador                 | 42 | 0.9524 | 0.9344 | 0.2619 | 0.2143 | 0.1818 | 0.0000 |
| Guatemala                   | 34 | 1.0000 | 1.0000 | 0.3824 | 0.3824 | 0.0000 | 0.0000 |
| Japan                       | 45 | 0.9556 | 0.9515 | 0.6444 | 0.6444 | 0.0345 | 0.0625 |
| Hong                        | 22 | 0.9545 | 0.9394 | 0.7273 | 0.7727 | 0.0000 | 0.1667 |
| France                      | 19 | 0.7895 | 0.6042 | 0.7895 | 0.8947 | 0.0667 | 0.7500 |
| Holand-Netherlands          | 16 | 0.9375 | 0.4839 | 1.0000 | 0.9375 | 0.0625 | 0      |
| Ecuador                     | 12 | 1.0000 | 1.0000 | 0.4167 | 0.4167 | 0.0000 | 0.0000 |
| Cambodia                    | 12 | 0.6667 | 0.6250 | 0.7500 | 0.5833 | 0.3333 | 0.3333 |

### Detailed Group Metrics: Education

| Group        | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------|-------|----------|----------|-----------|----------------|--------|--------|
| HS-grad      | 5209  | 0.8792   | 0.8654   | 0.3513    | 0.3281         | 0.2049 | 0.0752 |
| Bachelors    | 3530  | 0.8841   | 0.8692   | 0.6569    | 0.6810         | 0.0699 | 0.2040 |
| Some-college | 2919  | 0.8571   | 0.8028   | 0.2487    | 0.2264         | 0.3320 | 0.0803 |
| 11th         | 464   | 0.9483   | 0.8096   | 0.0927    | 0.0539         | 0.4884 | 0.0071 |
| Assoc-voc    | 952   | 0.8834   | 0.8754   | 0.6166    | 0.6366         | 0.0784 | 0.1781 |
| Masters      | 1216  | 0.8890   | 0.8424   | 0.7689    | 0.7747         | 0.0684 | 0.2527 |
| Prof-school  | 700   | 0.9371   | 0.8025   | 0.9043    | 0.9214         | 0.0253 | 0.4179 |
| Assoc-acdm   | 598   | 0.8645   | 0.8637   | 0.5201    | 0.5585         | 0.0932 | 0.1812 |
| Preschool    | 547   | 0.9909   | 0.9444   | 0.9616    | 0.9525         | 0.0095 | 0.0000 |
| 5th-6th      | 134   | 0.9403   | 0.9297   | 0.3284    | 0.2836         | 0.1591 | 0.0111 |
| Doctorate    | 1176  | 0.9660   | 0.7586   | 0.9549    | 0.9719         | 0.0089 | 0.5660 |

|         |     |        |        |        |        |        |        |
|---------|-----|--------|--------|--------|--------|--------|--------|
| 12th    | 183 | 0.9290 | 0.8931 | 0.2295 | 0.1913 | 0.2381 | 0.0213 |
| 9th     | 244 | 0.9303 | 0.9033 | 0.2582 | 0.2131 | 0.2222 | 0.0166 |
| 1st-4th | 72  | 0.9722 | 0.9699 | 0.3750 | 0.3472 | 0.0741 | 0.0000 |
| 7th-8th | 280 | 0.9429 | 0.8977 | 0.1893 | 0.1464 | 0.2642 | 0.0088 |
| 10th    | 354 | 0.9492 | 0.7819 | 0.0763 | 0.0480 | 0.5185 | 0.0122 |

### Detailed Group Metrics: Race + Sex

| Group                       | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|-----------------------------|-------|----------|----------|-----------|----------------|--------|--------|
| White + Male                | 11117 | 0.8674   | 0.8656   | 0.5581    | 0.5573         | 0.1194 | 0.1492 |
| White + Female              | 4555  | 0.9398   | 0.9350   | 0.3701    | 0.3570         | 0.0991 | 0.0373 |
| Black + Male                | 898   | 0.9120   | 0.9110   | 0.4510    | 0.4432         | 0.1062 | 0.0730 |
| Other + Male                | 378   | 0.9656   | 0.9324   | 0.8598    | 0.8413         | 0.0308 | 0.0566 |
| Asian-Pac-Islander + Female | 186   | 0.9140   | 0.9073   | 0.3817    | 0.3495         | 0.1549 | 0.0435 |
| Asian-Pac-Islander + Male   | 377   | 0.8302   | 0.8244   | 0.5809    | 0.6021         | 0.1279 | 0.2278 |
| Black + Female              | 735   | 0.9646   | 0.9525   | 0.2531    | 0.2422         | 0.0914 | 0.0164 |
| Amer-Indian-Eskimo + Female | 62    | 0.9194   | 0.8816   | 0.2419    | 0.1935         | 0.2667 | 0.0213 |
| Other + Female              | 174   | 0.9655   | 0.9322   | 0.8621    | 0.8391         | 0.0333 | 0.0417 |
| Amer-Indian-Eskimo + Male   | 96    | 0.8750   | 0.8545   | 0.2917    | 0.3333         | 0.1429 | 0.1176 |

### Detailed Group Metrics: Age + Sex

| Group              | Count | Accuracy | F1 Score | Base Rate | Selection Rate | FNR    | FPR    |
|--------------------|-------|----------|----------|-----------|----------------|--------|--------|
| Middle-Aged + Male | 5968  | 0.8686   | 0.8329   | 0.7202    | 0.7423         | 0.0758 | 0.2743 |
| Senior + Female    | 1208  | 0.9346   | 0.9345   | 0.4901    | 0.4710         | 0.0861 | 0.0455 |
| Young + Female     | 2227  | 0.9582   | 0.8839   | 0.1105    | 0.0894         | 0.2846 | 0.0116 |
| Young + Male       | 3785  | 0.8933   | 0.8322   | 0.2159    | 0.1810         | 0.3280 | 0.0458 |

|                      |      |        |        |        |        |        |        |
|----------------------|------|--------|--------|--------|--------|--------|--------|
| Middle-Aged + Female | 2277 | 0.9319 | 0.9311 | 0.5578 | 0.5529 | 0.0654 | 0.0715 |
| Senior + Male        | 3113 | 0.8542 | 0.8370 | 0.6637 | 0.6605 | 0.1123 | 0.2120 |

## Comparative Group Metrics: Before vs After (SMOTE)

### Age

| Group       | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Middle-Aged | 0.7746      | 0.8686     | 0.3347       | 0.0735      | 0.1281       | 0.1980      | 0.3208             | 0.6900            |
| Young       | 0.7674      | 0.8477     | 0.5238       | 0.3180      | 0.0218       | 0.0321      | 0.0731             | 0.1470            |
| Senior      | 0.7678      | 0.8702     | 0.3887       | 0.1065      | 0.0992       | 0.1503      | 0.2648             | 0.6075            |

### Marital-status

| Group                 | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-----------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Never-married         | 0.7551      | 0.8319     | 0.6257       | 0.4552      | 0.0023       | 0.0039      | 0.0187             | 0.0414            |
| Married-civ-spouse    | 0.7307      | 0.8154     | 0.3501       | 0.0951      | 0.1920       | 0.2826      | 0.3975             | 0.7271            |
| Divorced              | 0.7357      | 0.9015     | 0.6273       | 0.2165      | 0.0099       | 0.0210      | 0.0446             | 0.2159            |
| Separated             | 0.7656      | 0.8697     | 0.5862       | 0.3281      | 0.0058       | 0.0116      | 0.0375             | 0.1152            |
| Married-spouse-absent | 0.7764      | 0.9258     | 0.5556       | 0.0286      | 0.0075       | 0.1343      | 0.0592             | 0.7553            |
| Widowed               | 0.7088      | 0.7024     | 0.7097       | 0.6897      | 0.0000       | 0.0091      | 0.0251             | 0.0336            |
| Married-AF-spouse     | 0.7068      | 0.8970     | 0.6000       | 0.0121      | 0.0000       | 0.0000      | 0.1538             | 0.9645            |

### Occupation

| Group           | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-----------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Other-service   | 0.6179      | 0.9491     | 0.8364       | 0.0679      | 0.0050       | 0.0319      | 0.0120             | 0.2981            |
| Farming-fishing | 0.6990      | 0.9292     | 0.6087       | 0.0509      | 0.0432       | 0.0926      | 0.0865             | 0.6168            |
| Sales           | 0.7745      | 0.8579     | 0.4022       | 0.1784      | 0.0792       | 0.1079      | 0.2149             | 0.3961            |
| Exec-managerial | 0.8001      | 0.8458     | 0.2403       | 0.0965      | 0.1582       | 0.2118      | 0.4626             | 0.6870            |
| Craft-repair    | 0.6870      | 0.8421     | 0.5636       | 0.1804      | 0.0920       | 0.1363      | 0.1682             | 0.4636            |

|                   |        |        |        |        |        |        |        |        |
|-------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Prof-specialty    | 0.8072 | 0.8666 | 0.2582 | 0.0905 | 0.1331 | 0.1790 | 0.3948 | 0.6491 |
| Machine-op-inspct | 0.6223 | 0.9166 | 0.7685 | 0.0825 | 0.0339 | 0.0840 | 0.0633 | 0.4836 |
| Tech-support      | 0.7685 | 0.8800 | 0.4370 | 0.1626 | 0.0655 | 0.0800 | 0.2157 | 0.4376 |
| Adm-clerical      | 0.7707 | 0.8843 | 0.4415 | 0.1700 | 0.0473 | 0.0620 | 0.1154 | 0.2694 |
| Missing           | 0.7269 | 0.8697 | 0.6087 | 0.1901 | 0.0201 | 0.0542 | 0.0559 | 0.1904 |
| Handlers-cleaners | 0.7264 | 0.9550 | 0.6389 | 0.0500 | 0.0105 | 0.0397 | 0.0350 | 0.5051 |
| Transport-moving  | 0.7032 | 0.7072 | 0.5648 | 0.5205 | 0.0684 | 0.0940 | 0.1372 | 0.1857 |
| Protective-serv   | 0.7181 | 0.8515 | 0.4328 | 0.0779 | 0.1420 | 0.2037 | 0.2664 | 0.7591 |
| Priv-house-serv   | 0.8949 | 0.9778 | 0.3333 | 0.0065 | 0.0000 | 0.0204 | 0.0385 | 0.8998 |
| Armed-Forces      | 1.0000 | 0.5895 | 0      | 0.0412 | 0.0000 | 0.0000 | 0.0000 | 0.9538 |

### Relationship

| Group          | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|----------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Own-child      | 0.7707      | 0.9745     | 0.6250       | 0.0637      | 0.0000       | 0.0037      | 0.0047             | 0.1372            |
| Husband        | 0.7289      | 0.8154     | 0.3516       | 0.0897      | 0.1938       | 0.2890      | 0.3997             | 0.7420            |
| Not-in-family  | 0.7484      | 0.8775     | 0.6097       | 0.2831      | 0.0081       | 0.0194      | 0.0457             | 0.1693            |
| Unmarried      | 0.7135      | 0.9340     | 0.6941       | 0.1593      | 0.0033       | 0.0106      | 0.0229             | 0.1716            |
| Wife           | 0.7357      | 0.7702     | 0.3168       | 0.2086      | 0.2138       | 0.2516      | 0.4259             | 0.5344            |
| Other-relative | 0.6685      | 0.9571     | 0.7857       | 0.0665      | 0.0000       | 0.0207      | 0.0085             | 0.4618            |

### Race

| Group              | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| White              | 0.7868      | 0.8885     | 0.3850       | 0.1151      | 0.0729       | 0.1079      | 0.2110             | 0.4991            |
| Black              | 0.7960      | 0.9300     | 0.4653       | 0.1015      | 0.0211       | 0.0432      | 0.0835             | 0.3527            |
| Amer-Indian-Eskimo | 0.7697      | 0.8652     | 0.3636       | 0.1860      | 0.0522       | 0.0783      | 0.1032             | 0.2785            |
| Asian-Pac-Islander | 0.7802      | 0.8577     | 0.4062       | 0.1345      | 0.0696       | 0.1502      | 0.2060             | 0.5187            |

|       |        |        |        |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|
| Other | 0.7525 | 0.9323 | 0.5455 | 0.0316 | 0.0260 | 0.0519 | 0.0795 | 0.8406 |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|

## Sex

| Group  | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Female | 0.7967      | 0.9380     | 0.4614       | 0.0968      | 0.0208       | 0.0341      | 0.0786             | 0.3549            |
| Male   | 0.7756      | 0.8707     | 0.3771       | 0.1150      | 0.0952       | 0.1435      | 0.2550             | 0.5574            |

## Native-country

| Group         | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|---------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| United-States | 0.7894      | 0.8905     | 0.3841       | 0.1160      | 0.0692       | 0.1032      | 0.2020             | 0.4859            |
| Ireland       | 0.8125      | 0.8717     | 0.4000       | 0.0741      | 0.0000       | 0.1429      | 0.2500             | 0.7647            |
| Cuba          | 0.8137      | 0.8879     | 0.3750       | 0.1333      | 0.0455       | 0.0909      | 0.2000             | 0.4054            |
| Missing       | 0.8208      | 0.8157     | 0.3333       | 0.2687      | 0.0592       | 0.1053      | 0.2249             | 0.2968            |
| Puerto-Rico   | 0.9231      | 0.9597     | 0.2500       | 0.0545      | 0.0000       | 0.0222      | 0.0612             | 0.5300            |
| Philippines   | 0.7349      | 0.8661     | 0.5000       | 0.1111      | 0.0702       | 0.1579      | 0.1644             | 0.5417            |
| China         | 0.8454      | 0.7798     | 0.3000       | 0.2000      | 0.0455       | 0.2273      | 0.2500             | 0.4595            |
| Germany       | 0.7974      | 0.8875     | 0.3571       | 0.1250      | 0.0750       | 0.1000      | 0.2222             | 0.4875            |
| Mexico        | 0.6026      | 0.9262     | 0.8333       | 0.1489      | 0.0098       | 0.0195      | 0.0142             | 0.1746            |
| Peru          | 1.0000      | 1.0000     | 0.0000       | 0.0000      | 0.0000       | 0.0000      | 0.0909             | 0.7872            |
| India         | 0.7402      | 0.7815     | 0.4118       | 0.1667      | 0.1053       | 0.2632      | 0.3333             | 0.6364            |
| Guatemala     | 1.0000      | 1.0000     | 0            | 0.0000      | 0.0000       | 0.0000      | 0.0000             | 0.3824            |
| Taiwan        | 0.8661      | 0.9003     | 0.1429       | 0.0303      | 0.1250       | 0.1250      | 0.4667             | 0.8784            |
| Vietnam       | 0.8217      | 0.8389     | 0.0000       | 0.0000      | 0.0455       | 0.1818      | 0.0870             | 0.3793            |
| Jamaica       | 0.4468      | 0.9443     | 1.0000       | 0.0857      | 0.0000       | 0.0000      | 0.0000             | 0.5714            |
| Hungary       | 1.0000      | 0.8083     | 0.0000       | 0.0500      | 0.0000       | 0.3333      | 0.2500             | 0.8696            |
| Portugal      | 0.4118      | 0.8956     | 1.0000       | 0.0625      | 0.0000       | 0.0000      | 0.0000             | 0.8182            |
| Greece        | 0.5625      | 0.8303     | 0.6250       | 0.0455      | 0.1667       | 0.3333      | 0.2857             | 0.8214            |
| Haiti         | 0.8857      | 0.9099     | 0.3333       | 0.0625      | 0.0000       | 0.1176      | 0.1000             | 0.6531            |
| Nicaragua     | 1.0000      | 0.9544     | 0            | 0.0690      | 0.0000       | 0.0000      | 0.0000             | 0.5870            |
| South         | 0.8462      | 0.9637     | 0.2857       | 0.0200      | 0.0526       | 0.0526      | 0.2308             | 0.7246            |
| France        | 0.4167      | 0.6042     | 0.3333       | 0.0667      | 0.7500       | 0.7500      | 0.7143             | 0.8947            |

|                                    |        |        |        |        |        |        |        |        |
|------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Canada                             | 0.6485 | 0.9165 | 0.6471 | 0.1034 | 0.0645 | 0.0645 | 0.1667 | 0.4667 |
| El-Salvador                        | 0.8177 | 0.9344 | 0.5000 | 0.1818 | 0.0000 | 0.0000 | 0.0571 | 0.2143 |
| England                            | 0.7039 | 0.8586 | 0.5385 | 0.2400 | 0.0800 | 0.0400 | 0.2105 | 0.4000 |
| Poland                             | 0.5567 | 0.9734 | 0.8333 | 0.0000 | 0.0000 | 0.0833 | 0.0556 | 0.8254 |
| Outlying-US<br>(Guam-USV<br>I-etc) | 1.0000 | 1.0000 | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.7353 |
| Honduras                           | 0.4706 | 1.0000 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.7037 |
| Dominican-<br>Republic             | 0.8857 | 0.8730 | 0.0000 | 0.1429 | 0.0556 | 0.1111 | 0.1500 | 0.4375 |
| Italy                              | 0.7375 | 0.9089 | 0.5000 | 0.1111 | 0.0556 | 0.0556 | 0.2143 | 0.5556 |
| Trinidad&T<br>obago                | 1.0000 | 0.8523 | 0.0000 | 0.0385 | 0.0000 | 0.2222 | 0.1000 | 0.8851 |
| Iran                               | 0.5758 | 0.7727 | 0.6667 | 0.1471 | 0.1818 | 0.2727 | 0.2143 | 0.7111 |
| Columbia                           | 0.8190 | 1.0000 | 0.0000 | 0.0000 | 0.0556 | 0.0000 | 0.1053 | 0.4375 |
| Hong                               | 0.4167 | 0.9394 | 1.0000 | 0.0000 | 0.1667 | 0.1667 | 0.1429 | 0.7727 |
| Yugoslavia                         | 0.4167 | 0.2500 | 1.0000 | 1.0000 | 0.0000 | 0.6000 | 0.0000 | 0.5000 |
| Ecuador                            | 1.0000 | 1.0000 | 0      | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.4167 |
| Cambodia                           | 0.5333 | 0.6250 | 0.7500 | 0.3333 | 0.0000 | 0.3333 | 0.1429 | 0.5833 |
| Japan                              | 0.8392 | 0.9515 | 0.2857 | 0.0345 | 0.0625 | 0.0625 | 0.2609 | 0.6444 |
| Scotland                           | 0.4444 | 0.9396 | 1.0000 | 0.0192 | 0.0000 | 0.0000 | 0.0000 | 0.9107 |
| Laos                               | 0.7778 | 0.9372 | 0.5000 | 0.0286 | 0.0000 | 0.0000 | 0.1667 | 0.8718 |
| Thailand                           | 1.0000 | 0.9292 | 0      | 0.0167 | 0.0000 | 0.1250 | 0.0000 | 0.8824 |

## Education

| Group        | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|--------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| Bachelors    | 0.7999      | 0.8692     | 0.2605       | 0.0699      | 0.1445       | 0.2040      | 0.3916             | 0.6810            |
| 9th          | 0.4918      | 0.9033     | 1.0000       | 0.2222      | 0.0000       | 0.0166      | 0.0000             | 0.2131            |
| HS-grad      | 0.6998      | 0.8654     | 0.6203       | 0.2049      | 0.0408       | 0.0752      | 0.0942             | 0.3281            |
| Assoc-voc    | 0.7245      | 0.8754     | 0.4924       | 0.0784      | 0.0904       | 0.1781      | 0.2012             | 0.6366            |
| Prof-school  | 0.8304      | 0.8025     | 0.0318       | 0.0253      | 0.3582       | 0.4179      | 0.7857             | 0.9214            |
| 10th         | 0.6942      | 0.7819     | 0.7143       | 0.5185      | 0.0061       | 0.0122      | 0.0282             | 0.0480            |
| Some-college | 0.7408      | 0.8028     | 0.4918       | 0.3320      | 0.0638       | 0.0803      | 0.1450             | 0.2264            |

|                |        |        |        |        |        |        |        |        |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|
| 7th-8th        | 0.5622 | 0.8977 | 0.9091 | 0.2642 | 0.0000 | 0.0088 | 0.0080 | 0.1464 |
| Masters        | 0.7977 | 0.8424 | 0.2107 | 0.0684 | 0.1886 | 0.2527 | 0.5243 | 0.7747 |
| 11th           | 0.6970 | 0.8096 | 0.7143 | 0.4884 | 0.0048 | 0.0071 | 0.0181 | 0.0539 |
| 5th-6th        | 0.4762 | 0.9297 | 1.0000 | 0.1591 | 0.0000 | 0.0111 | 0.0000 | 0.2836 |
| Assoc-acd<br>m | 0.7488 | 0.8637 | 0.4674 | 0.0932 | 0.0732 | 0.1812 | 0.1847 | 0.5585 |
| Preschool      | 1.0000 | 0.9444 | 0.0000 | 0.0095 | 0.0000 | 0.0000 | 0.0455 | 0.9525 |
| 12th           | 0.7640 | 0.8931 | 0.6154 | 0.2381 | 0.0000 | 0.0213 | 0.0325 | 0.1913 |
| Doctorate      | 0.6641 | 0.7586 | 0.1569 | 0.0089 | 0.5283 | 0.5660 | 0.7355 | 0.9719 |
| 1st-4th        | 0.4945 | 0.9699 | 1.0000 | 0.0741 | 0.0000 | 0.0000 | 0.0000 | 0.3472 |

### Race + Sex

| Group                       | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-----------------------------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
| White + Female              | 0.7997      | 0.9350     | 0.4439       | 0.0991      | 0.0244       | 0.0373      | 0.0883             | 0.3570            |
| Black + Male                | 0.7823      | 0.9110     | 0.4766       | 0.1062      | 0.0304       | 0.0730      | 0.1183             | 0.4432            |
| White + Male                | 0.7732      | 0.8656     | 0.3673       | 0.1194      | 0.1056       | 0.1492      | 0.2721             | 0.5573            |
| Black + Female              | 0.7903      | 0.9525     | 0.4865       | 0.0914      | 0.0128       | 0.0164      | 0.0444             | 0.2422            |
| Amer-Indian-Eskimo + Male   | 0.7630      | 0.8545     | 0.2000       | 0.1429      | 0.0735       | 0.1176      | 0.1233             | 0.3333            |
| Asian-Pac-Islander + Female | 0.7707      | 0.9073     | 0.5625       | 0.1549      | 0.0087       | 0.0435      | 0.0611             | 0.3495            |
| Amer-Indian-Eskimo + Female | 0.6964      | 0.8816     | 0.6667       | 0.2667      | 0.0213       | 0.0213      | 0.0566             | 0.1935            |
| Asian-Pac-Islander + Male   | 0.7469      | 0.8244     | 0.3875       | 0.1279      | 0.1329       | 0.2278      | 0.2941             | 0.6021            |
| Other + Female              | 0.7292      | 0.9322     | 0.5000       | 0.0333      | 0.0417       | 0.0417      | 0.0769             | 0.8391            |
| Other + Male                | 0.6992      | 0.9324     | 0.6667       | 0.0308      | 0.0189       | 0.0566      | 0.0645             | 0.8413            |

### Age + Sex

| Group | F1 (Before) | F1 (After) | FNR (Before) | FNR (After) | FPR (Before) | FPR (After) | Sel. Rate (Before) | Sel. Rate (After) |
|-------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|
|-------|-------------|------------|--------------|-------------|--------------|-------------|--------------------|-------------------|

|                      |        |        |        |        |        |        |        |        |
|----------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Middle-Aged + Female | 0.8162 | 0.9311 | 0.3745 | 0.0654 | 0.0407 | 0.0715 | 0.1514 | 0.5529 |
| Young + Male         | 0.7590 | 0.8322 | 0.5221 | 0.3280 | 0.0307 | 0.0458 | 0.0949 | 0.1810 |
| Senior + Male        | 0.7519 | 0.8370 | 0.3512 | 0.1123 | 0.1538 | 0.2120 | 0.3536 | 0.6605 |
| Young + Female       | 0.7672 | 0.8839 | 0.5455 | 0.2846 | 0.0116 | 0.0116 | 0.0393 | 0.0894 |
| Middle-Aged + Male   | 0.7460 | 0.8329 | 0.3273 | 0.0758 | 0.1850 | 0.2743 | 0.3950 | 0.7423 |
| Senior + Female      | 0.7571 | 0.9345 | 0.5349 | 0.0861 | 0.0260 | 0.0455 | 0.0798 | 0.4710 |

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Report generated by Dataset Fairness Evaluation System